

Lab 16 - Microsoft Sentinel & Entra ID Cloud Security Monitoring

Cloud Identity Logging, KQL Hunting, and Audit Investigation

Objective: Build and validate a cloud security monitoring workflow using Microsoft Sentinel and Microsoft Entra ID. The lab connects Entra audit activity to a Log Analytics workspace, generates a controlled identity-management event, and investigates that activity using Kusto Query Language (KQL).

Environment

Component	Purpose
Microsoft Azure	Cloud platform used to host the lab resources
Log Analytics workspace	Stores and queries Sentinel/Entra log data
Microsoft Sentinel	Cloud-native SIEM used for monitoring and investigation
Microsoft Entra ID	Identity and access platform generating audit activity
Microsoft Entra ID connector	Forwards supported Entra logs into Sentinel
KQL	Queries and filters Entra audit events for analyst investigation

Workspace Deployment

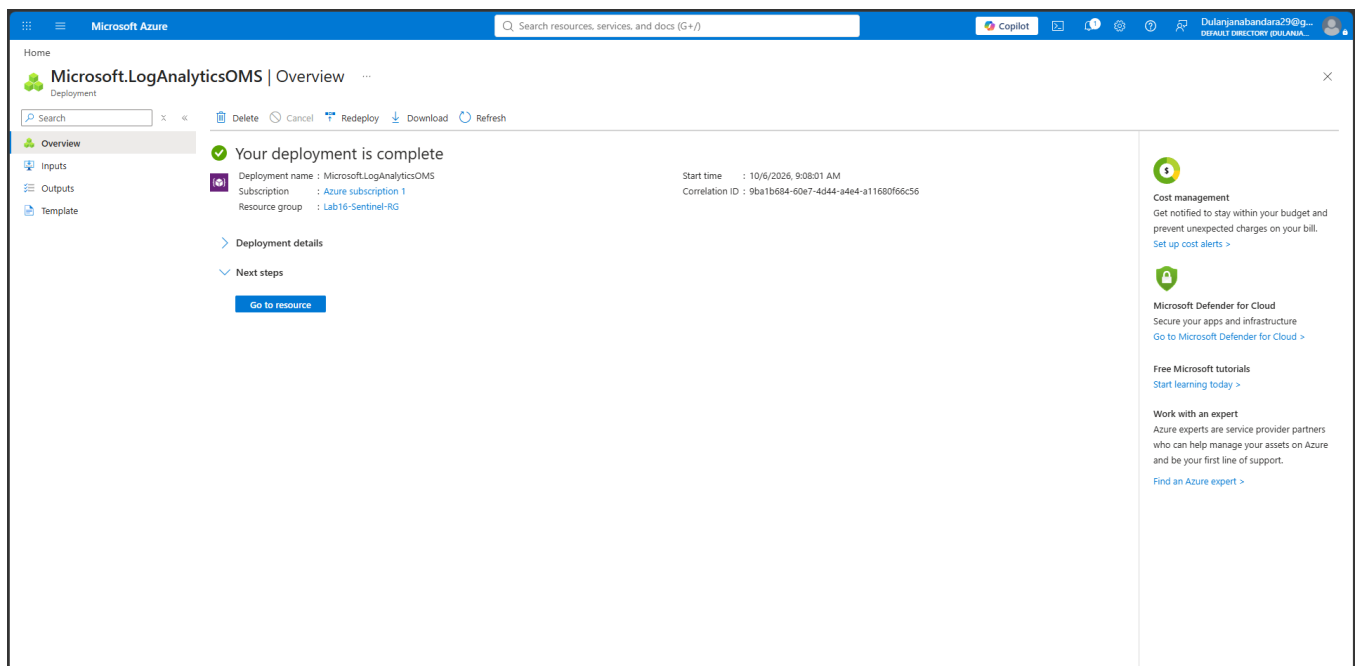


Figure 1 - Log Analytics workspace deployment completed successfully for Lab 16.

Microsoft Entra ID Solution Installation

The Microsoft Entra ID solution was installed from the Sentinel Content Hub so that the Entra data connector and related Sentinel content became available to the workspace.

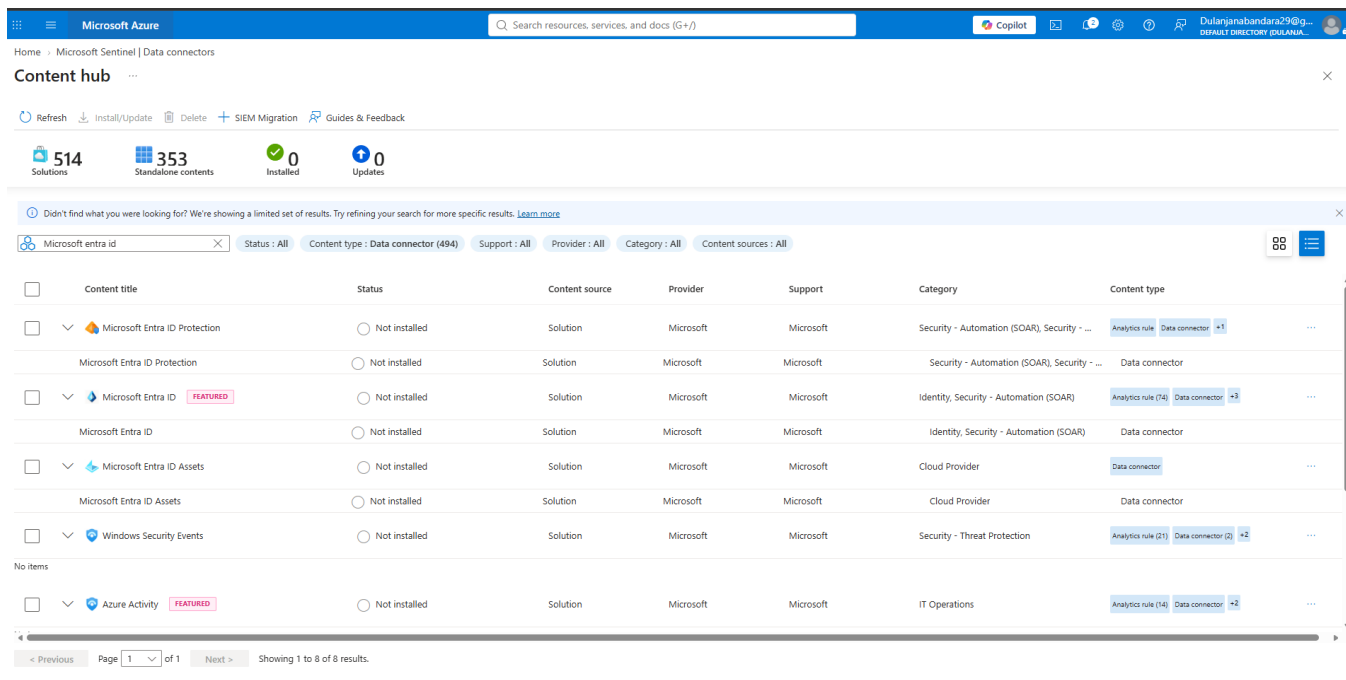


Figure 2 - Microsoft Entra ID solution visible in the Sentinel Content Hub.

Connector Configuration

The Microsoft Entra ID connector showed that **Audit Logs** could be connected in the current tenant. The portal also displayed that **Sign-In Logs** require Microsoft Entra ID P1 or P2 licensing, so this lab used AuditLogs for the practical investigation.

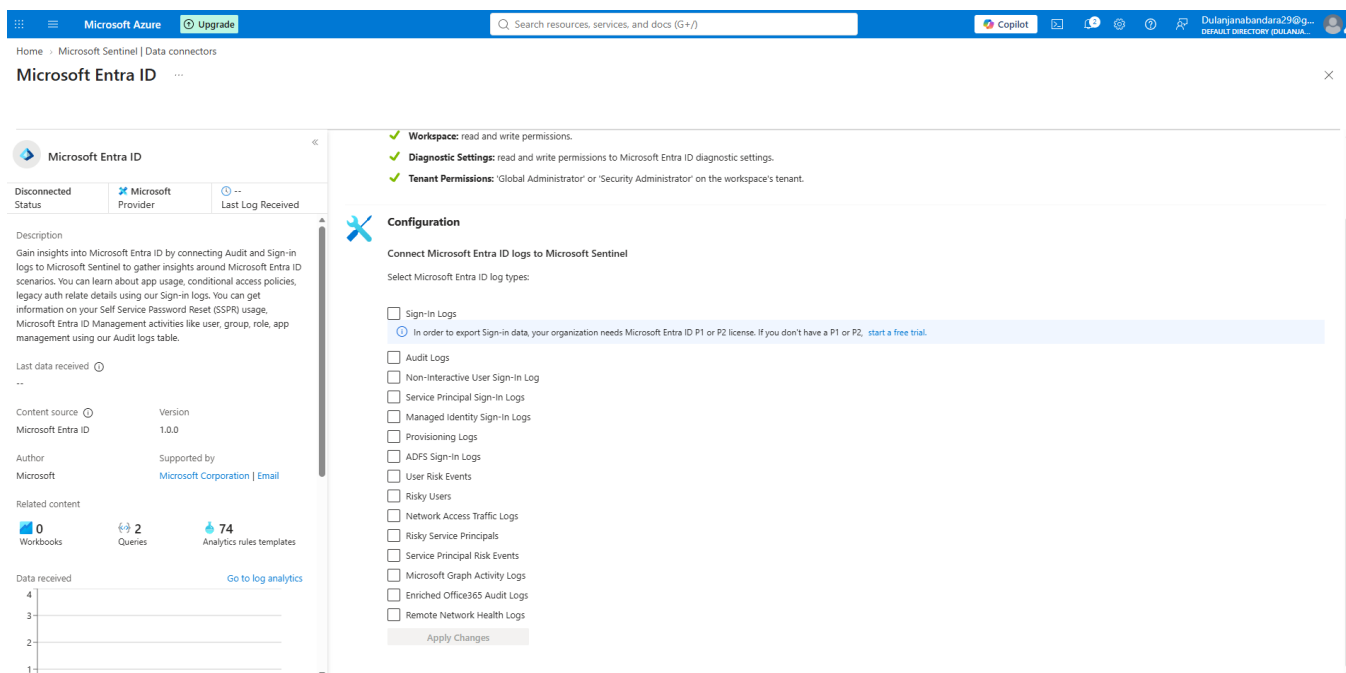


Figure 3 - Entra connector page showing Audit Logs and the P1/P2 requirement for Sign-In Logs.

Controlled Identity Event

A security group named **Lab16-Test-Group** was created in Microsoft Entra ID to generate a controlled, benign identity-management audit event. After ingestion into the Log Analytics workspace, the event became available in the **AuditLogs** table.

Initial KQL Validation

```
AuditLogs
| sort by TimeGenerated desc
```

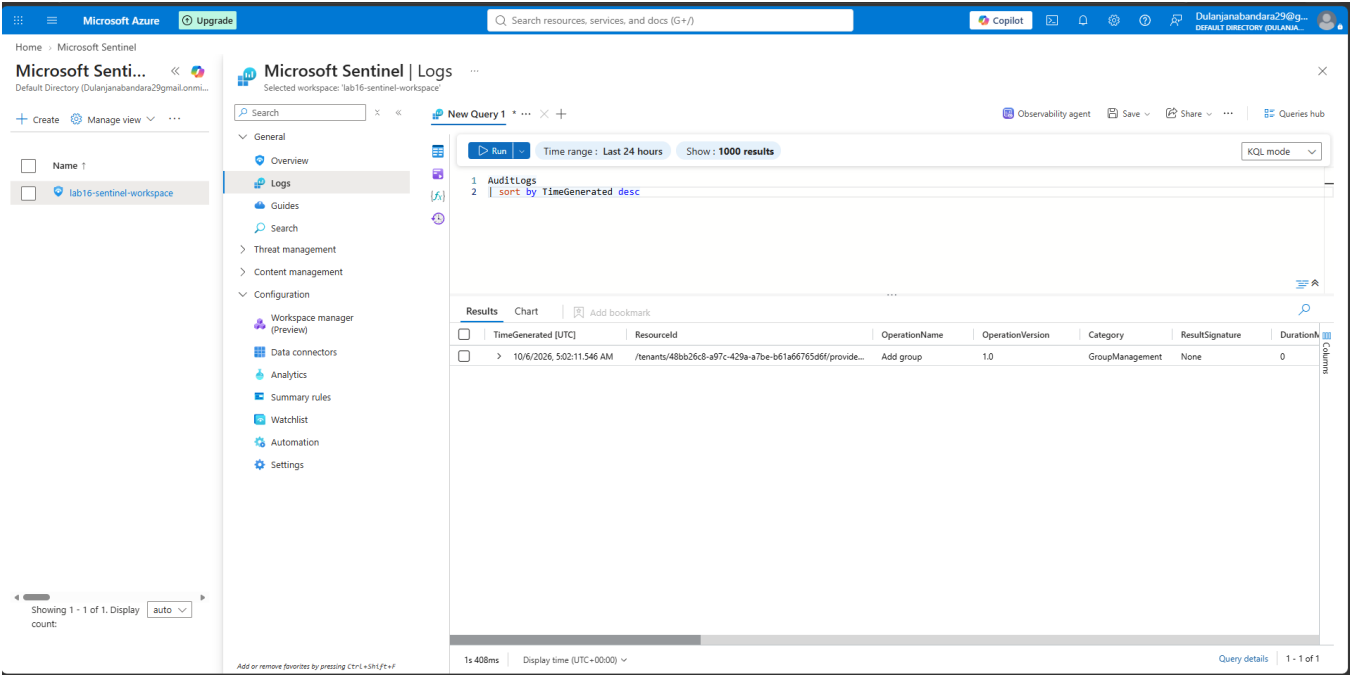


Figure 4 - Initial AuditLogs query returning the real Entra group-creation event in Microsoft Sentinel.

Field	Observed value
OperationName	Add group
Category	GroupManagement
Result	success
Target resource	Lab16-Test-Group
Log source	Microsoft Entra ID AuditLogs

Focused KQL Investigation

A more focused KQL query was used to isolate group-creation events and project the most useful analyst fields, including the initiator and target resource.

```
AuditLogs
| where OperationName == "Add group"
| project TimeGenerated, OperationName, Category, Result, InitiatedBy, TargetResources
| sort by TimeGenerated desc
```

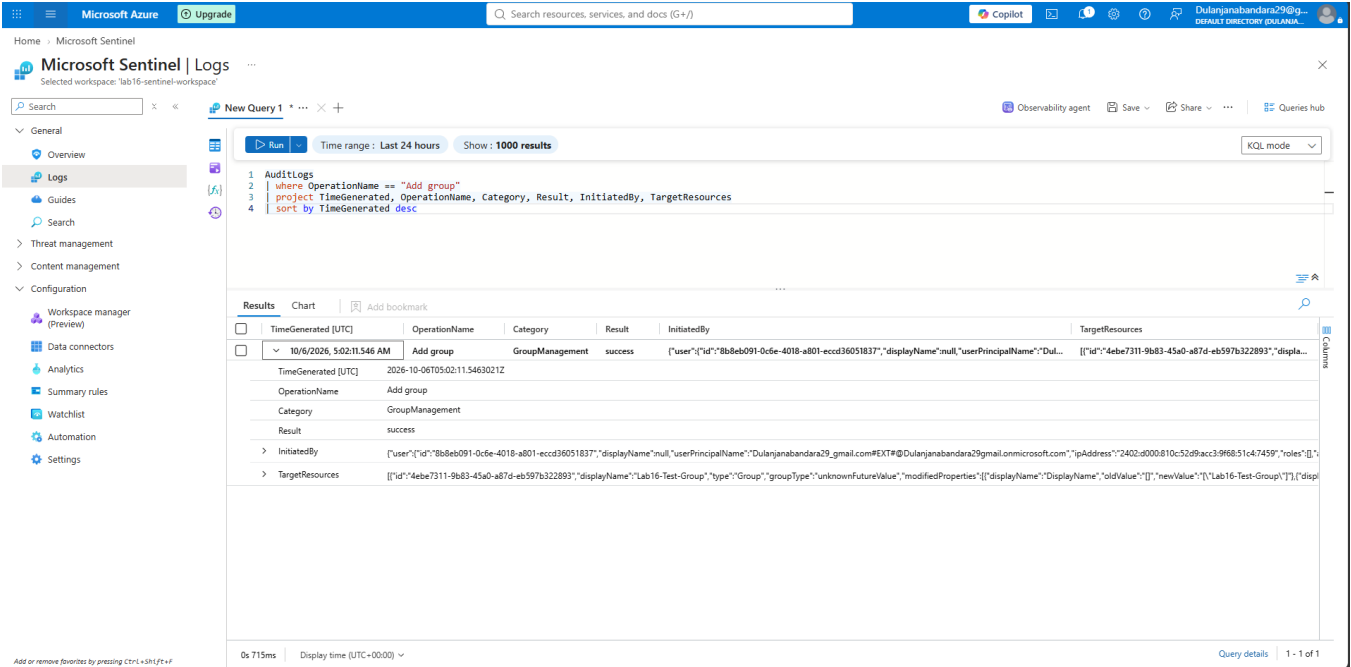


Figure 5 - Focused Add group query showing success, initiator context, and target resource details.

Analyst Interpretation

The event was successfully traced to a known, authorized lab action. The record showed a successful GroupManagement operation and included both the initiating Entra user context and the target group. In a production environment, similar events could be reviewed for unauthorized group creation, privilege changes, or unexpected identity administration.

Broader Identity-Management Hunt

The final hunt expanded the scope from a single operation to recent user and group management activity. This demonstrates how an analyst can move from a known event to a broader identity-change investigation.

```
AuditLogs
| where Category in ("UserManagement", "GroupManagement")
| project TimeGenerated, OperationName, Category, Result, InitiatedBy, TargetResources
| sort by TimeGenerated desc
```

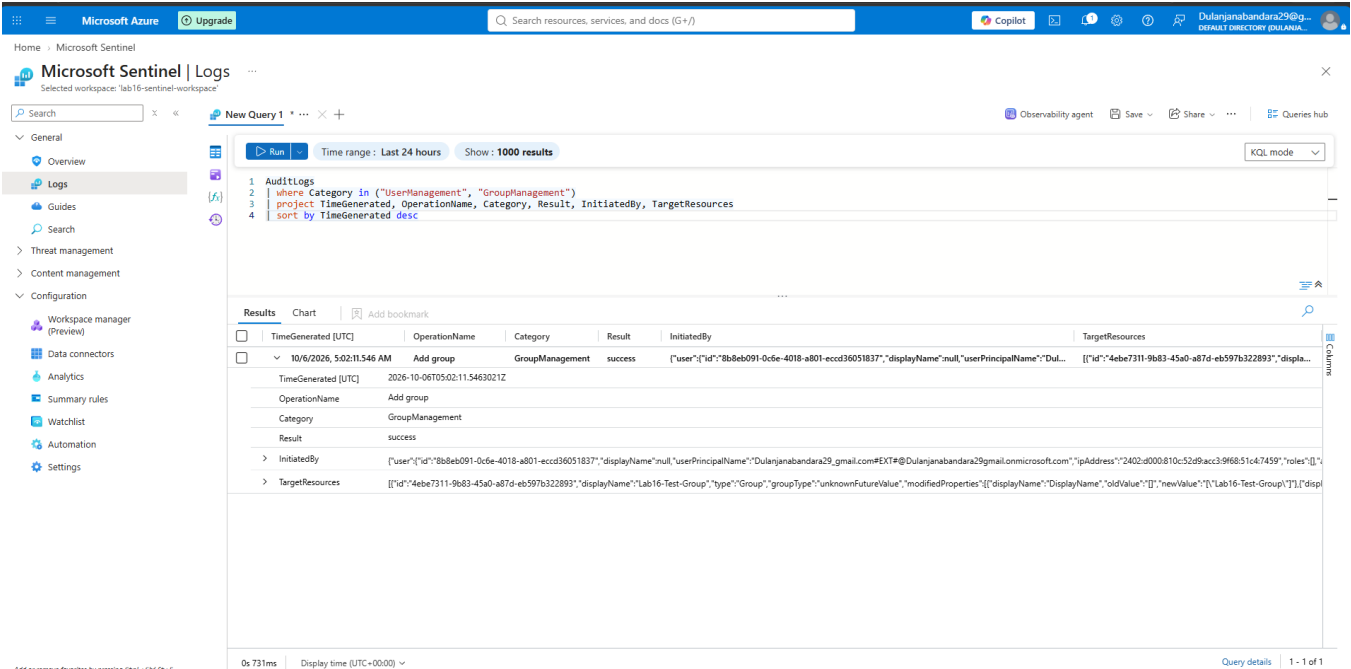


Figure 6 - Broader KQL hunt across UserManagement and GroupManagement audit activity.

Hunting objective	Result
Identity log ingestion	Successful
Group-management activity	Detected
Initiator context	Available
Target-resource context	Available
KQL filtering and projection	Validated

Cloud Security Engineering Assessment

Capability	Assessment
Cloud SIEM setup	Microsoft Sentinel enabled on a dedicated Log Analytics workspace
Identity data onboarding	Microsoft Entra ID Audit Logs connected successfully
IAM / identity monitoring	Group-management activity observed and investigated
KQL	Queries used for sorting, filtering, projecting, and hunting audit events
Investigation context	InitiatedBy and TargetResources fields reviewed
Controlled validation	Benign Lab16-Test-Group creation used to generate evidence
Licensing limitation	Sign-In Logs not enabled because the tenant did not have Entra ID P1/P2

Cloud Security Lessons

- Identity activity is a critical cloud-security telemetry source because user, group, role, and application changes can affect access across the environment.
- Microsoft Sentinel uses Log Analytics as the data layer for querying and investigating connected security telemetry.
- KQL allows analysts and security engineers to move from broad log review to focused hunting with filters and projected fields.
- AuditLogs can provide useful identity-change evidence even when premium SignInLogs are unavailable.
- Initiator and target-resource context are important when determining whether an identity-management action is expected or suspicious.
- Cloud-security investigations should document licensing and telemetry limitations so conclusions are not overstated.

Final Conclusion

Lab 16 successfully demonstrated a cloud identity-monitoring workflow using Microsoft Azure, Microsoft Sentinel, Log Analytics, Microsoft Entra ID, and KQL. Entra audit activity was connected to Sentinel, a controlled security-group creation event was generated, and the event was identified and investigated through both focused and broader KQL queries. The lab provides hands-on evidence of cloud SIEM onboarding, identity and access monitoring, Entra audit analysis, KQL hunting, and analyst interpretation of cloud identity-management activity.

Safety note: All activity was performed in an authorized personal lab environment using a benign test group. No production tenant or third-party identity environment was modified. Sign-In Logs were not used because the current tenant did not include the required Microsoft Entra ID P1/P2 licensing.